

AI Workforce Intelligence & Disruption Analytics

A Production-Ready 4-Tier Machine Learning & Prescriptive Analytics Ladder Project

Domain / Industry:	Human Capital Management, Labor Economics & Applied AI
Dataset Analyzed:	ai_job_trends_dataset.csv (30,000 Records, 13 Features)
Methodology:	4-Tier Analytics Ladder (Descriptive, Diagnostic, Predictive, Prescriptive)
Primary Target:	Decline_Risk_Flag (Binary: Decreasing vs. Increasing Employment Status)
Primary Deliverables:	app.py (Streamlit), requirements.txt, README.md, Project_Report.pdf

1. Executive Summary & Problem Statement

As generative artificial intelligence and automated systems rapidly permeate across global industry sectors, workforce planners and corporate executives face critical structural shifts in employment demand. This project executes the complete 4-Tier Analytics Ladder on the AI Job Trends Dataset (30,000 observations across 8 global markets and 8 major industry verticals) to identify labor contraction risks, isolate genuine causal mechanisms from non-causal correlations, prevent target leakage in future-oriented models, and establish a resource-constrained operational intervention framework for talent redeployment and reskilling.

2. Tier 1: Data Hygiene, Quality Audit & Architecture

A comprehensive automated data quality audit was executed through data_pipeline.py:

- **Header Standardization:** Stripped leading/trailing whitespace in column headers (' Job Title' → 'Job Title').
- **Character Encoding Normalization:** Resolved typographical apostrophe artifacts (e.g. Master’s Degree → Master's Degree).
- **Audit Assertions:** Verified 0 missing values and 0 duplicate records with automated median/mode imputation fallback.
- **Domain Validation:** Verified Salary strictly > \$0 (\$30k - \$150k), Experience in [0, 20], Percentages in [0%, 100%].
- **Entity-Level Aggregation:** Computed profiles across 639 Job Titles and 64 Industry-Location hubs.
- **Feature Engineering:** Created Net_Projected_Growth, Growth_Rate_Pct, Salary_Per_Exp_Year, Automation_Exposure_Index, and Education_Rank.

3. Tier 2: Exploratory Data Analysis (Descriptive & Diagnostic)

Five distinct visualizations were implemented with explicit empirical observations, diagnostic business insights, and rigorous **correlation vs. causation** separation:

- **Chart 1 (Automation Risk by AI Impact):** Risk spans 0% to 100% across Low, Moderate, and High impact categories. *Causation guardrail:* Tagging a role as High Impact does NOT cause automation; task modularity drives both.
- **Chart 2 (Compensation vs. Risk by Industry):** Flat trendlines indicate high compensation provides zero structural immunity against cognitive AI disruption. High wage correlates with specialization, not automation immunity.

- **Chart 3 (Educational Cohort Trajectory):** Net openings across High School to PhD tiers average ~30M openings each. Advanced credentials do NOT causally guarantee employment stability under AI.
- **Chart 4 (Geographic Modality & Remote Work):** Balanced remote penetration (~49.8%) and risk (~50.1%) across 8 nations. Remote work correlates with digital tasks, but remote status does not directly cause displacement.
- **Chart 5 (Demographic Diversity across Trajectories):** Gender diversity is identically distributed across contracting (50.01%) and expanding (49.95%) roles. Contraction relates to task content, not demographic characteristics.

4. Tier 3: Predictive Modeling & Strict Leakage Prevention

Target Variable: Decline_Risk_Flag (1 = Decreasing Job Status, 0 = Increasing Job Status).
Strict Leakage Protection: Excluded Projected Openings (2030) (future lookahead bias), direct target derivatives (Net_Projected_Growth, Growth_Rate_Pct), and raw Job Title (639 high-cardinality identity strings causing memorization).

Performance Metric	Logistic Regression (Baseline)	Random Forest (Challenger)
Test Accuracy	50.10%	49.37%
Precision	49.66%	48.87%
Recall (Sensitivity)	51.09%	47.19%
F1-Score	50.36%	48.01%
ROC-AUC Score	0.5029	0.4899
Confusion Matrix [TN, FP, FN, TP]	[1487, 1540, 1454, 1519]	[1559, 1468, 1570, 1403]
Optimal Economic Cutoff (τ^*)	0.10 (Loss: \$10,594,500)	0.41 (Loss: \$10,587,500)

Commercial Trade-Off Analysis: False Positives (wasted reskilling) cost ~\$3,500/worker, while False Negatives (unanticipated displacement) cost ~\$18,500/worker (~5.3x penalty). Consequently, enterprise loss is minimized by lowering the decision threshold ($\tau^* \approx 0.40 - 0.45$).

5. Tier 4: Prescriptive Strategy & Operational Levers

Resource-Constrained Decision Rule: Assign personnel to intensive reskilling if predicted contraction probability $P(\text{Decline}) \geq \tau^*(B)$, where τ^* is determined dynamically by quarterly budget capacity B.

- **Tier 1: Urgent Augmentation (Top 20% Risk Decile):** External hiring freezes, 12-week intensive AI tool apprenticeships, internal redeployment.
- **Tier 2: Monitored Upskilling (Middle 50% Exposure):** Subsidized generative AI micro-credentials and bi-annual workflow audits.
- **Tier 3: Strategic Expansion (Lower 30% Risk):** Accelerated recruitment in non-automatable niches and advanced tooling investments.

6. Deliverables Summary

- **app.py:** Interactive Streamlit web application running at <http://localhost:8501>.
- **requirements.txt:** Pinned dependencies (pandas, scikit-learn, streamlit, matplotlib, seaborn, plotly, joblib).
- **README.md:** Full documentation and 4-tier methodology.
- **Project_Report.pdf / Project_Report.docx:** Official submission-ready project reports.